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Assignment No. 1

Assignment on :-

Conditional statements (if, elif, else),

Loops (for, while),

Loop control statements (break, continue, pass),

Boolean logic and truth tables,

Problem-Solving Session: Practice loop/condition problems.

Conditional Statement

1. Largest among Three Numbers: Write a Python program that takes three numbers as input and prints out the largest among them.

```
a = float(input("Please enter the number: "))
b = float(input("Please enter the number: "))
c = float(input("Please enter the number: "))

print ("The Largest number is:", max (a,b,c))
```

or

```
num1 = float (input("Please enter the number: "))
num2 = float (input("Please enter the number: "))
num3 = float (input("Please enter the number: "))

if num1 >= num2 and num1 >= num3:
    print ("The largest number is num1")

elif num2 >= num1 and num2 >= num3:
    print ("The largest number is num2")

else:
    print ("The largesrnumber is num3")
```

2. Quadrant Identifier: Write a Python program that takes the coordinates (x, y) of a point as input and prints out which quadrant it belongs to (1st, 2nd, 3rd, or 4th).

```
x = float(input("Enter x coordinate: "))
y = float(input("Enter y coordinate: "))

if x > 0 and y > 0:
    print ("The point is in the 1st Quadrant")

elif x > 0 and y < 0:
    print ("The point is in the 2nd Quadrant")

elif x < 0 and y < 0:
    print ("The point is in the 3rd Quadrant")

elif x < 0 and y > 0:
    print ("The point is in the 4th Quadrant")
```

```

elif x == 0 and y != 0:
    print ("The point lies on the Y-axis")

elif x != 0 and y == 0:
    print ("The point lies on the X-axis")

else:
    print ("The point is in the (0,0)")

```

3. Age Classifier: Write a Python program that takes a person's age as input and prints out whether they are an infant (0-1), toddler (2-3), child (4-12), teenager (13-19), adult (20+).

```

age = float(input("Please enter your age: "))

if age < 0:
    print ('Invalid')

elif age <= 0 and age <= 1:
    print('infant')

elif age >= 2 and age <= 3:
    print ('Toddler')

elif age >= 4 and age <= 12:
    print ('Child')

elif age >= 13 and age <= 19:
    print ('Teenager')

else:
    print('Adult')

```

4. BMI Calculator: Write a Python program that takes a person's height (in meters) and weight (in kilograms) as input and calculates their Body Mass Index (BMI). Print out their BMI and a message indicating whether they are underweight (<18.5), normal (18.5-24.9), overweight (25-29.9), or obese (>=30).

```

weight = float (input("Enter your weight in kilogram: "))
height = float (input("Enter your height in meter: "))

bmi = weight / (height ** 2)

```

```

print (f"Your weight is {bmi:.2f}")

if bmi < 18.5:
    print ("You are under weight")

elif 18.5 <= bmi <= 24.9:
    print ("You are normal weight")

elif 25 <= bmi <= 29.9:
    print ("You are over weight")

else:
    print ("You are obse")

```

5. Temperature Converter: Write a Python program that takes a temperature input in Celsius and converts it to Fahrenheit if the temperature is above 0°C, or to Kelvin if the temperature is below 0°C.

```

celsius = float(input("Enter temperature in Fahrenheit: "))

if celsius > 0:
    fahrenheit = (celsius *9/5) + 32
    print ("Temperature in Fahrenheit:", fahrenheit)

elif celsius < 0:
    kalvin = celsius + 273.15
    print ("Temperature in Kelvin:", kalvin)

else:
    print ("Temperature is exactly 0'C. Temperature in neither above nor below 0'C.")

```

6. Simple Calculator: Input two numbers and an operator (+, -, *, /). Use if-elif to perform the operation and print the result. Handle division by zero using conditions.

```

num1 = float (input("Please enter first number: "))
num2 = float (input("Please enter second number: "))

op = input("Enter operator(+, -, *, /): ")

if op == "+":
    result = num1 + num2
    print ("Result:", result)

```

```

elif op == "-":
    result = num1 - num2
    print ("Result:", result)

elif op == "*":
    result = num1 * num2
    print ("Result:", result)

elif op == "/":
    if num2 == 0:
        print ("Error, cannot be divided by Zero")
    else:
        result = num1 / num2
        print ("Result:", result)

else:
    print ("Invalid Operator")

```

Loops

1. FizzBuzz: Write a Python program that prints the numbers from 1 to 100. But for multiples of three, print "Fizz" instead of the number, and for the multiples of five, print "Buzz". For numbers that are multiples of both three and five, print "FizzBuzz" using a for loop.

```

for i in range (1, 101):

    if i % 3 == 0 and i % 5 == 0:
        print ("FizzBuzz")

    elif i % 3 == 0:
        print ("Fizz")

    elif i % 5 == 0:
        print ("Buzz")

    else:
        print (i)

```

2. Armstrong Number Checker: Write a Python program that takes an integer input from the user and checks if it is an Armstrong number (a number that is equal to the sum of its own digits raised to the power of the number of digits)

using a for loop.

3. Number Reverser: Write a Python program that takes an integer input from the user and prints its reverse using a while loop.

```
num = int(input("Enter an integer: "))

reverse_num = 0

while num > 0:
    digit = num % 10
    reverse_num = reverse_num * 10 + digit
    num = num // 10

print("Reversed number:", reverse_num)
```

4. Sum of Digits: Write a Python program that takes an integer input from the user and calculates the sum of its digits using a while loop.

```
num = int(input("Enter an integer: "))

temp = abs(num)
sum_digits = 0

while temp > 0:
    digits = temp % 10
    sum_digits += digits
    temp //= 10

print("Sum of digits:", sum_digits)
```

5. Print the pattern (for/while):

1

22

333

4444

55555

Using for loop:-

```
for i in range (1,6):
    print (str(i) * i)
```

Using while loop:-

```
i = 1
while i <= 5:
    print (str(i) * i)

    i +=1
```

6. Fibonacci Sequence (for loop): Input N. Print the first N terms of the Fibonacci sequence.

```
N = int (input("Enter how many Fibonacci you want to print: "))

a, b = 0, 1

for i in range (N):
    print (a, end= " ")

    a, b = b, b + a
```

7. Password Retry System (while loop) : Predefined password. Let user try until correct or after 3 failed attempts show "Account locked"

```
Password = '1234'
Attempts = 0

while Attempts < 3:
    user_input = input('Please enter Password: ')

    if user_input == Password:
        print ('Access Granted')
        break
    else:
        Attempts += 1
        print ('Wrong Password, Attempt Left,', 3 - Attempts)

    if Attempts == 3:
        print ('Account Locked')
```

8. Guess the Number Game (while loop): Consider a random number between 1 and 10. Let user guess until correct.